

Neural PK-PD Modeling with ODE for Early Drug Discovery

Master's diploma thesis presentation draft

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Problem Context

- Early drug discovery requires simultaneous assessment of **efficacy**, **safety**, **absorption**, and **elimination**.
- Existing studies typically optimize for one or two ADMET endpoints in isolation.
- PK-PD modeling and molecular ML are rarely connected within a single reproducible workflow.
- Calibration is usually an afterthought, yet uncalibrated probabilities distort downstream risk simulation.

Thesis Goal and Research Gap

- Four ADME-Tox endpoints predicted jointly:
 - Binding affinity (pIC50)
 - hERG inhibition (cardiotoxicity)
 - Caco-2 permeability (absorption)
 - Hepatic clearance
- Single reproducible pipeline: from raw molecules to clinical trial simulation.
- Bridge benchmark ML and **translational decision support**.

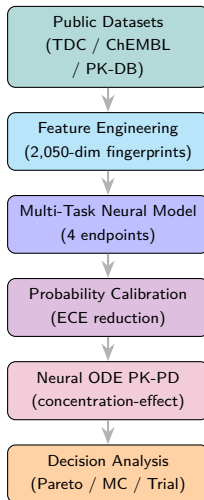


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Four Principal Objectives

- 1 Build a **leakage-controlled multi-task dataset** from heterogeneous public sources (TDC, ChEMBL, PK-DB).
- 2 Develop predictive models for **binding, hERG, Caco-2, and clearance** with explicit evaluation of real-data integration and GNN fusion.
- 3 Integrate **calibrated model outputs** with Neural ODE PK-PD simulation for mechanistic analysis.
- 4 Evaluate translational behavior via **Pareto trade-off analysis**, Monte Carlo robustness, and virtual clinical trial simulation.

Key Contributions

- **Full experimental pipeline:** data curation → feature engineering → multi-task training → calibration → PK-PD simulation → decision analysis.
- **Thesis-essential modeling sequence:** TDC fingerprint upgrade → real ChEMBL binding integration → hERG refinement → joint GNN+MLP fusion.
- **Decision-support layer:** Pareto efficacy-safety optimization, Monte Carlo uncertainty quantification, and in-silico clinical trial OC analysis.
- **Exploratory RapidDock prototype:** distance-biased transformer for geometry-aware protein-ligand modeling (feasibility study).

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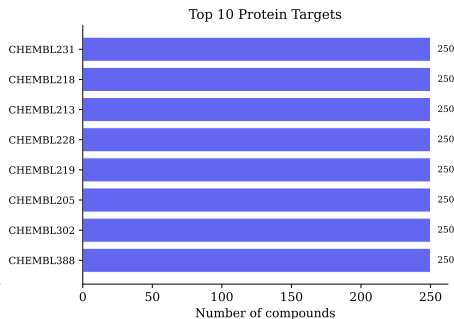
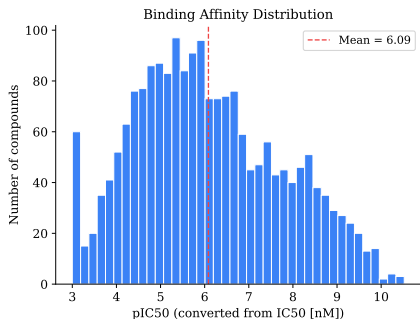
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Data Sources

Source	Rows	Role
TDC (hERG / Caco-2 / clearance)	2,768	Core ADMET benchmark training labels
ChEMBL (real-SMILES binding)	3,410	Pharmacodynamic potency (pIC50)
PK-DB	796	Mechanistic ODE calibration context
PubChem	5,000	Illustrative safety screening reference

- **4,768** harmonized rows, **4 tasks**, **2,050 features**.
- Split 70 / 10 / 20; stratified for hERG & Caco-2.
- Quality gate: MD5 deduplication, leakage audit, K-S distribution check.

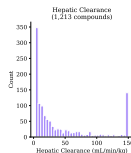
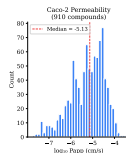
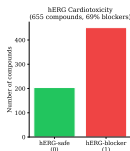
Data Sources – ChEMBL Binding Affinity



pIC50 histogram across 2,000 compounds (left) and top-10 pharmacological target coverage by protein class (right). Mean pIC50 = 6.09.

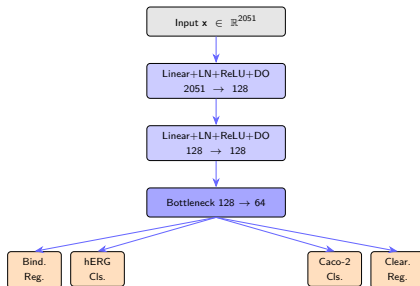
Feature Engineering

- **Morgan FP:** 2,048 bits, radius 2 (RDKit).
- **Descriptors:** MW, logP.
- $\mathbf{x} \in \mathbb{R}^{2050}$.
- Task sample counts:
 - Binding: 2,000
 - hERG: 655
 - Caco-2: 900
 - Clearance: 1,213



TDC endpoint distributions: hERG class balance (left), Caco-2 permeability (centre), hepatocyte clearance (right).

Multi-Task Model Architecture

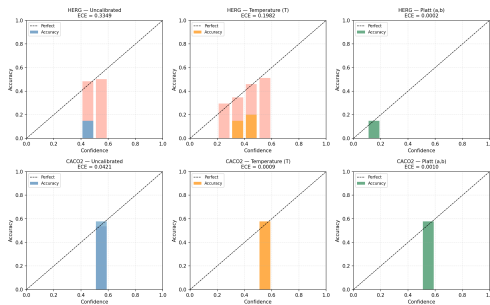


- **Shared encoder:** LayerNorm preferred over BatchNorm for interleaved task batches.
- **Heads from 64-dim bottleneck:**
 - Bind: $64 \rightarrow 64 \rightarrow 32 \rightarrow 1$ (deep reg.)
 - hERG/Caco-2: $64 \rightarrow 32 \rightarrow 1$ (logits)
 - Clearance: $64 \rightarrow 32 \rightarrow 1$ (reg.)
- **Loss:** focal ($\gamma=2$) + MSE + physics penalty $\mathcal{P}$.

Calibration and Reliability

- High AUROC $\neq$ reliable probabilities.
- hERG & Caco-2 probabilities feed *directly* into PK-PD as risk variables / k_a .
- **Temp. scaling** (hERG):
 $\hat{p} = \sigma(z/T)$.
- **Platt scaling** (Caco-2):
 $\hat{p} = \sigma(az+b)$.

Section 18: Reliability Diagrams — hERG & Caco-2 Calibration



Reliability diagrams: uncalibrated (left), temperature-scaled (centre), Platt-scaled (right). Top row: hERG; bottom row: Caco-2. Diagonal = perfect calibration.

Task	ECE_{before}	ECE_{after}
hERG (temp.)	0.1727	0.0058
Caco-2 (Platt)	0.3583	0.0448

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Endpoint-to-ODE Parameter Mapping

Predicted endpoint	PK/PD role	Parameter
Clearance (regression)	Elimination rate	$k_e = CL/V_c$
Caco-2 probability (classif.)	Oral absorption rate	k_a
Binding pChEMBL (regression)	PD potency	$EC_{50} = 10^{-pChEMBL}$
hERG probability (classif.)	Cardiac safety flag	Safety threshold

Two-compartment PK ODE:

$$\frac{dC_c}{dt} = -(k_e + k_{12})C_c + k_{21}C_t + u(t), \quad \frac{dC_t}{dt} = k_{12}C_c - k_{21}C_t$$

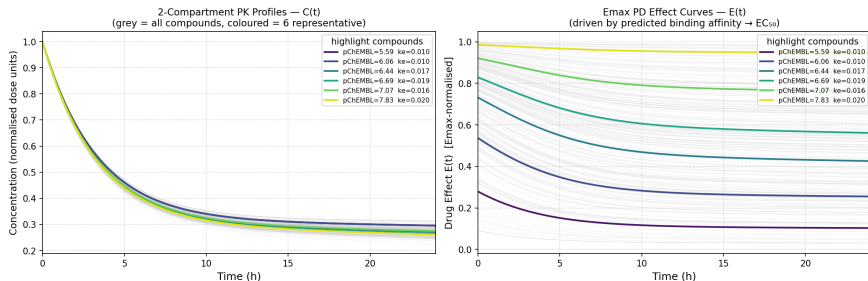
E_{max} PD model:

$$E(t) = E_{\max} \cdot \frac{C_c(t)}{EC_{50} + C_c(t)}$$

Solver: `torchdiffeq.odeint` (Dormand-Prince RK45).

Neural ODE PK-PD Simulation Results

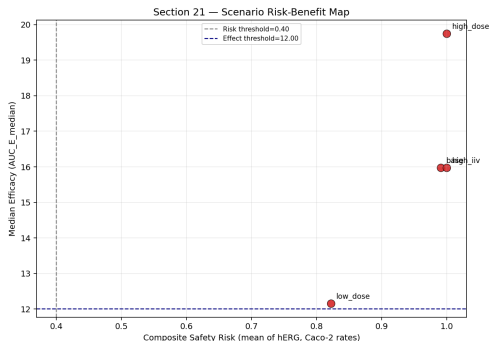
Section 16: Neural ODE PK/PD Simulation (n=200 compounds, model_15 predictions)



Left: 2-compartment PK concentration profiles $C(t)$ for 200 compounds (6 highlighted by pChEMBL and k_e). Right: Emax PD effect curves $E(t)$ driven by predicted binding affinity $\rightarrow EC_{50}$. Higher pChEMBL $\Rightarrow$ higher sustained effect.

Decision-Layer Analyses

- **Pareto front:** efficacy vs. safety trade-off is genuine—no single dominant regimen.
- **Knee-point:** most balanced point on Pareto frontier.
- **Monte Carlo** ($n=1000$): validates regimen robustness; CIs narrow.
- **Virtual trial:** clear dose-response in efficacy; safety burden persists.



Scenario risk-benefit map: all regimens exceed safety risk threshold (0.40) while only high-dose regimens surpass efficacy target (12.0).
Confirms multi-objective tension.

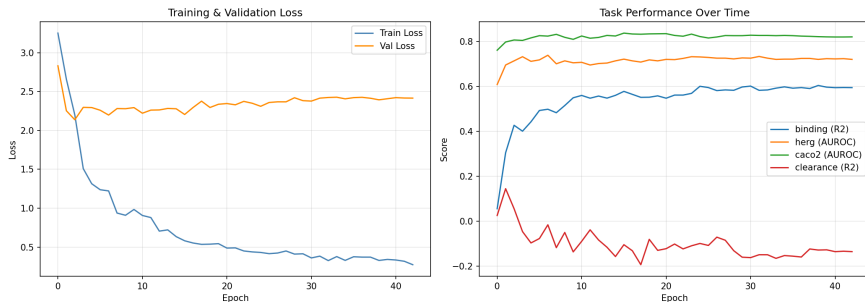
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Experimental Progression

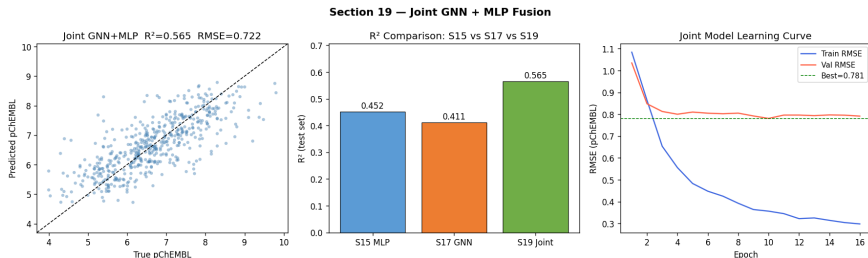
Stage	Key finding
1. Baseline (256-bit FP)	Near-zero binding R^2 ; hERG AUROC $\approx$ random
2. 2048-bit real SMILES	Structural signal enters TDC tasks; clear AUROC lift
3. Real ChEMBL binding	Largest binding R^2 gain; confirms data-realism bottleneck
4. hERG refinement	hidden_dim 256 + pos_weight sweep $\rightarrow$ AUROC > 0.80
5. Calibration	ECE_{hERG} : 0.17 $\rightarrow$ 0.006; $ECE_{\text{Caco-2}}$: 0.36 $\rightarrow$ 0.045
6. GNN+MLP fusion	Strongest binding R^2 ; graph+FP representations complementary
7–8. PK-PD + Pareto	Genuine efficacy-safety trade-off; knee-point regimen selected
9. Monte Carlo	Regimen robustness confirmed; bootstrap CIs narrow
10. Virtual trial	Drug recommendation in efficient 6 trials

Multi-Task Training Dynamics



Left: training and validation loss converge stably across 43 epochs. Right: per-task performance over time—hERG (orange) and Caco-2 (green) AUROC plateau early; binding R^2 (blue) grows steadily; clearance R^2 (red) improves after initial instability.

GNN+MLP Fusion: Binding Affinity Results



Left: joint GNN+MLP parity plot ($R^2=0.565$, RMSE= 0.722). Centre: R^2 comparison—MLP baseline (S15: 0.452), GNN alone (S17: 0.411), joint fusion (S19: 0.565). Right: joint model learning curve with best val RMSE= 0.781.

Target Performance Metrics

Task	Metric	Target
Binding affinity (regression)	R^2	> 0.60
hERG inhibition (classification)	AUROC	> 0.80
Caco-2 permeability (classification)	AUROC	> 0.75
Hepatic clearance (regression)	RMSE	< 1.0

- March 8 baseline reference: hERG AUROC = 0.4836, Caco-2 AUROC = 0.4719, binding $R^2 = 0.0019$, clearance RMSE = 0.9766.
- Leakage remediation (MD5 dedup + retrain from scratch on clean splits) was applied before reporting final results.

Limitations

- 1 **Public data quality:** assay heterogeneity (different protocols, species, cell lines) introduces label noise.
- 2 **ODE parameter assumptions:** k_{12} , k_{21} are fixed population averages, not per-compound predictions.
- 3 **GNN scope:** graph encoder trained only for the binding task; other three tasks retain MLP-only heads.
- 4 **No prospective validation:** virtual trial is a model-based planning exercise, not a substitute for empirical data.
- 5 **RapidDock:** exploratory prototype; no calibration, uncertainty quantification, or real PDB protein context.

Conclusions and Future Work

- Real-data integration and GNN+MLP fusion materially improve predictive quality.
- Post-hoc calibration is a **necessary prerequisite** for using classifier probabilities in mechanistic simulation.
- The integrated pipeline supports **multi-objective translational analysis**—Pareto, Monte Carlo, virtual trial—revealing that efficacy gains do not eliminate safety constraints.

Future work:

- ① External and prospective validation.
- ② Per-compound ODE parameter prediction (leverage PK-DB data).
- ③ Expanded GNN to all four tasks; extended RapidDock branch with real protein structures.
- ④ Interactive deployment as a decision-support tool.

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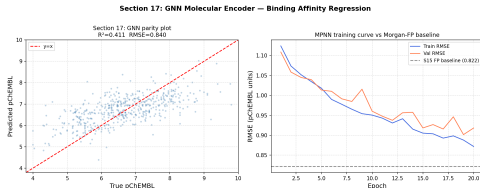
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GNN Architecture (Backup)

- **Atom features:** atomic number (one-hot), degree, formal charge, #H, aromaticity.
- **Bond features:** type, ring membership.
- **3-layer GCN:**

$$h_v^{(k)} = \text{ReLU}\left(W^{(k)} \cdot \text{MeanAGG}_{\mathcal{N}(v)}\right)$$

- **Fusion:** $z_{\text{fus}} = [z_{\text{GNN}}; z_{\text{MLP}}]$.



GNN binding parity plot ($R^2=0.411$) and MPNN training curve vs. S15 MLP fingerprint baseline (RMSE 0.822, dashed).

RapidDock Prototype (Backup)

- Input: ~ 80 – 120 protein-ligand complexes (140 ChEMBL SMILES, RDKit 3D embedding).

- **Distance-biased attention:**

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}} - 0.25 \cdot D_{\text{joint}}\right) V$$

- 4 transformer blocks (8 heads, $d_{\text{head}} = 16$), learnable type embeddings.
- Symmetry-aware distance loss over permutations of same-type atom groups.
- Ligand coordinate reconstruction via L-BFGS (200 steps).
- **Status:** feasibility prototype only—no calibration or real PDB structures.

Training Hyperparameters (Backup)

Hyperparameter	Value
Optimizer	Adam
Learning rate	10^{-3} (fine-tune: 3×10^{-4})
Weight decay	10^{-4}
Gradient clip	norm = 1.0
Batch size	64
Max epochs	300
Early stopping patience	40 (extended 60 for hERG sweep)
LR scheduler	ReduceLRonPlateau, factor=0.5, patience=10
Focal loss γ	2.0 (hERG, Caco-2)
Random seed	42

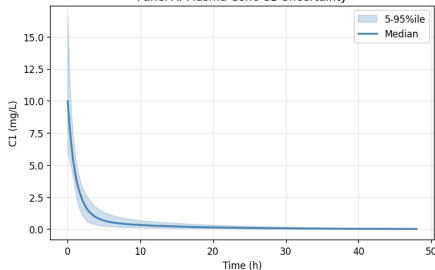
Software Stack (Backup)

- Python 3.14
- PyTorch 2.10.0 (MPS/CUDA; Neural ODE training)
- RDKit 2025.9.5 (fingerprints, SMILES, 3D conformers)
- torchdiffeq 0.2.5 (Dormand-Prince RK45 ODE solver)
- scikit-learn 1.8.0 (preprocessing, metrics, calibration)
- pandas 3.0.0, numpy 2.4.1
- Fixed seed (SEED=42) across NumPy, PyTorch, and Python random.

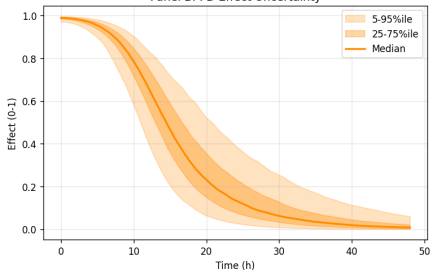
Population PK-PD Simulation (Backup)

Section 20 - Population PK-PD Simulation with Uncertainty Bounds

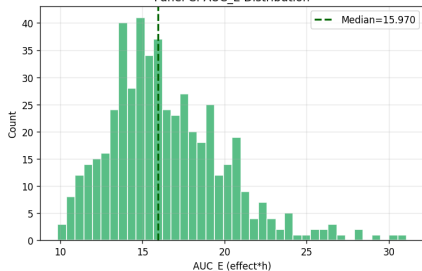
Panel A: Plasma Conc C1 Uncertainty



Panel B: PD Effect Uncertainty



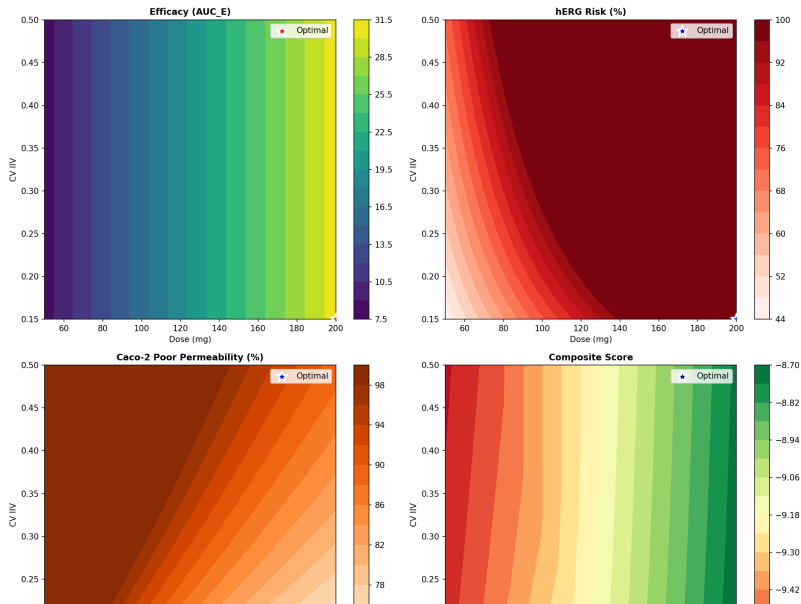
Panel C: AUC_E Distribution



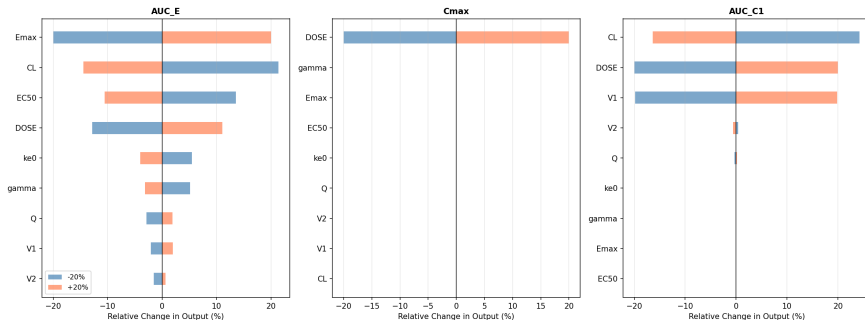
Panel D: Safety Flags (hERG & Caco-2)



Dose Optimization Grid (Backup)



Sensitivity Analysis – Tornado Plot (Backup)



Tornado (one-at-a-time $\pm 20\%$) sensitivity for three ODE outputs. Left: AUC_E most sensitive to Emax, CL, EC50. Centre: C_{max} dominated by dose. Right: AUC_{C1} dominated by CL and dose. Results confirm that predicted CL and binding affinity are the primary levers for PK-PD decisions.

Thank you